

Q1. Describe diagnosis, investigations and management of hypertension.

Or (essential hypertension and hypertensive retinopathy)

Or

Hypertensive patient in dental practice.

Q2. Ischemic Heart Disease (IHD)

Q3. Cushing Syndrome

Q4. Describe clinical features, complications, diagnosis and management of chronic rheumatic mitral stenosis.

Q5. Describe aetiology, clinical profile, investigations and management of acute myocardial infarction.

Q6. What are the symptoms and signs of congestive cardiac failure? How will you treat this condition.

Q7. Describe clinical features, investigations and treatment of Acute Rheumatic Fever (ARF)

Q8. Disseminated Intravascular Coagulation (DIC)

Q9. Name Anticoagulant Drugs, Uses, and Precautions in Dental Practice

Q10. Describe signs and symptoms of angina and how you would manage the case.

Q11. Discuss aetiology and clinical features of subacute bacterial endocarditis. Add a note on principles of management of Infective Endocarditis prophylaxis.

Q12. Cyanosis

Q13. Syncope (Fainting)

Q14. Cardiopulmonary resuscitation.

Q1. Describe diagnosis, investigations and management of hypertension.

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Hypertensive patient in dental practice.

Hypertension is defined as a level of blood pressure where the risk for **target organ damage** increases, and the benefits of treatment outweigh the costs and hazards. Around **15%** of the general population can be considered hypertensive.

Classification of Hypertension

I. Primary (Essential) Hypertension

- Elevated blood pressure without any identifiable cause.
- **85-90%** of cases.
- **70%** of patients have a **positive family history**.
- **Multifactorial** pathogenesis: high salt intake, heavy alcohol use, obesity, sedentary lifestyle.

II. Secondary Hypertension

- Elevated blood pressure due to a specific, identifiable cause.
- Accounts for **15%** of cases.
- Causes:
 - **Renal disorders:** Renovascular stenosis, glomerulonephritis, polycystic kidney disease.
 - **Endocrinal disorders:** Pheochromocytoma, Cushing's syndrome, primary hyperaldosteronism, acromegaly, hypo/hyperthyroidism.
 - **Drugs:** Oral contraceptives, corticosteroids, sympathomimetics, NSAIDs, cyclosporine.
 - **Miscellaneous:** Toxaemia of pregnancy, coarctation of aorta, raised intracranial pressure, obstructive sleep apnoea.

Classification of Blood Pressure in Adults

Category	Systolic BP (mmHg)	Diastolic BP (mmHg)
Normal	<120	<80
Prehypertension	120-139	80-89
Stage 1 Hypertension	140-159	90-99
Stage 2 Hypertension	>160	>100

Clinical Features

- **Asymptomatic** in most cases, diagnosed on routine exams.
- **Symptoms due to high BP:** Occipital headache, dizziness, palpitation, fatigue.
- **Symptoms of target organ damage:** Epistaxis, hematuria, blurred vision, TIA, angina, breathlessness.
- **Symptoms of underlying causes:**
 - Weight gain (Cushing's syndrome).
 - Weight loss (thyrotoxicosis).

- Episodic headache, sweating (pheochromocytoma).

Physical Examination Findings

- **Truncal obesity** (Cushing's disease).
- **Palpable kidneys** (polycystic kidney disease).
- **Radiofemoral delay** (coarctation of aorta).
- **Recurrent back pain, undiagnosed fever** (chronic pyelonephritis).
- **Abdominal bruit** (renovascular disease).
- **Signs of complications:**
 - Heaving apex, 4th heart sound, loud aortic second heart sound.
 - Pulmonary crackles, retinal changes.

Risk Factors

- **Smoking, obesity, hyperlipidaemia, diabetes mellitus, family history of premature CAD.**

Complications of Hypertension

1. CNS Complications

- **TIA, CVA (stroke), subarachnoid hemorrhage.**
- **Hypertensive encephalopathy:** Very high BP with neurological symptoms (speech/vision disturbances, fits, loss of consciousness, papilloedema). Symptoms are usually reversible with BP control.

2. Ophthalmic Complications

- **Hypertensive retinopathy:** Thickening of retinal arterioles, narrowing of blood vessels, arteriovenous nipping, retinal hemorrhages, exudates, papilloedema.
 - **Grading of Retinopathy:**
 - **Grade I:** Arteriolar narrowing, increased light reflex.
 - **Grade II:** Marked arteriolar narrowing, arteriovenous nipping.
 - **Grade III:** Grade II + hemorrhages, exudates.
 - **Grade IV:** Grade III + papilloedema.

3. Cardiovascular Complications

- **Coronary artery disease** (angina, myocardial infarction).
- **Left ventricular failure.**
- **Aortic aneurysm, aortic dissection.**

4. Renal Complications

- **Proteinuria, progressive renal failure.**

5. Malignant Hypertension

- **Markedly high BP** with retinal hemorrhages/exudates, confusion, headache, vomiting, visual disturbances, and renal function deterioration.

Investigations in Hypertension

Basic Investigations for All Patients

- **Urine analysis:** Check for **protein, blood, and glucose.**
- **Blood urea nitrogen and serum creatinine:** Assess **renal function.**
- **Serum electrolytes:** Check **sodium and potassium** levels.
- **Blood glucose:** **Fasting and post-prandial** to assess **hyperglycaemia.**
- **Serum lipid profile.**
- **Serum calcium and uric acid.**
- **ECG:** Look for **left ventricular hypertrophy.**
- **Chest X-ray:** Check **cardiac size, cardiac failure, and aortic dilatation.**

Special Group of Patients

- **Chest X-ray and echocardiography.**
- **Renal ultrasonography and angiography.**
- **Serum calcium and phosphate.**
- **TSH** (Thyroid-stimulating hormone).
- **Urinary cortisol and catecholamines.**
- **Plasma renin activity and aldosterone.**

Treatment of Hypertension

1. General Measures (Lifestyle Modifications)

- **Relief of stress:** Avoid unnecessary tension; practice relaxation techniques.
- **Salt restriction:** Low sodium diet (**<100 mEq sodium** or **<6 g salt/day**).
- **Weight reduction:** Caloric restriction in obese patients to reduce BP.
- **Control of risk factors:** Restrict cholesterol and saturated fats, reduce oil intake, stop smoking, and moderate alcohol consumption.
- **Regular exercises, relaxation classes, meditation, and biofeedback.**

2. Antihypertensive Drug Therapy

Various classes of drugs are used to treat hypertension:

A. Diuretics

- **Thiazide diuretics:**
 - Chlorothalidone, **Chlorothiazide**, **Hydrochlorothiazide**, **Metolazone**.
- **Loop diuretics:**
 - **Furosemide**, **Bumetanide**.
- **Potassium-sparing diuretics:**
 - **Amiloride**, **Triamterene**.
- **Aldosterone receptor blockers:**
 - **Spirolonolactone**.

B. Beta-blockers

- **Propranolol**, **Metoprolol**, **Atenolol**, **Bisoprolol**.

C. Combined Alpha- and Beta-blockers

- **Carvedilol**, **Labetalol**.

D. ACE Inhibitors

- **Captopril**, **Enalapril**, **Lisinopril**, **Ramipril**, **Perindopril**.

E. Angiotensin II Antagonists

- **Losartan**, **Candesartan**.

F. Calcium Channel Blockers

- **Nifedipine**, **Verapamil**, **Diltiazem**, **Amlodipine**.

G. Alpha Blockers

- **Prazosin**, **Terazosin**, **Doxazosin**.

H. Direct Vasodilators

- **Hydralazine**, **Minoxidil**.

I. Central α_2 -blockers and Other Central Agents

- **Clonidine**, **Alpha-methyldopa**, **Reserpine**.

Diuretics:

- **Thiazides** are the most commonly used diuretics.

- **Side effects** of thiazides can be minimized by using a **low dose** (e.g., **25 mg/day** of hydrochlorothiazide).
- A combination of **thiazides** and **potassium-sparing agents** (e.g., **spironolactone, amiloride**) can be used.
- **Loop diuretics** are used less often.

Beta-blockers:

- Lower **heart rate** and **cardiac output**.
- **Cardioselective** beta-blockers (**atenolol, metoprolol**) are preferred.
- Used with caution in **heart failure, heart blocks, and asthma**.
- **Contraindications: COPD, asthma, heart failure, heart block, peripheral vascular disease, diabetes mellitus.**

ACE Inhibitors and Angiotensin Receptor Blockers (ARBs):

- **ACE inhibitors** (e.g., **enalapril, lisinopril**) and **ARBs** (e.g., **losartan, irbesartan**) have **good antihypertensive effects**.
- **ARBs** cause **fewer side effects** (no **cough** or **angioedema**) and are better tolerated.
- Contraindicated in patients with **renal impairment** and **bilateral renal artery stenosis**.

Calcium Channel Blockers:

- Act by causing **arterial vasodilatation**.
- Used in **angina, hypertensive crisis, Raynaud's disease, migraine, bronchial asthma, pulmonary hypertension, and congestive cardiac failure**.
- Common drugs: **Nifedipine, verapamil**.

Other Drug Groups:

- **Alpha adrenergic receptor blockers:** Prazosin, doxazosin.
- **Centrally acting agents:** Clonidine, methyldopa.
- **Vasodilators:** Hydralazine, minoxidil.
- **Mixed alpha- and beta-blockers:** Labetalol, Carvedilol.

Broad Guidelines for Drug Selection:

- **Diuretics** (thiazides) are often the first choice.
- **Beta-blockers** are useful in **young hypertensives** and those with **ischemic heart disease**.
- **Calcium channel blockers** are ideal for **elderly patients**, those with **asthma**, and **peripheral vascular disease**.

- **ACE inhibitors** are beneficial in patients with **diabetes, heart failure, and renovascular hypertension.**

Guidelines for Treatment Based on Hypertension Stage:

Stage of Hypertension	Drug Treatment
Prehypertension (without risk factors)	None
Prehypertension (with risk factors)	Diuretics
Stage 1 Hypertension	Beta-blockers, ACE inhibitors, Calcium channel blockers, Angiotensin receptor blockers, Thiazide diuretics
Stage 2 Hypertension	Thiazide diuretics + Beta-blockers, ACE inhibitors, Calcium channel blockers, Angiotensin receptor blockers. Add other drugs if BP is not controlled.

Q2. Ischemic Heart Disease (IHD)

Definition

Ischemic heart disease (IHD) is a condition where there is a **reduced blood flow** to the **heart muscles**, leading to **insufficient oxygen supply**. This is often caused by **atherosclerosis** of the coronary arteries.

Causes Ischemic Heart Disease (IHD)

- **Atherosclerosis:** **Fatty plaques** build up inside the **coronary arteries**, narrowing them.
- **Thrombosis:** A **blood clot** forms over the plaque, blocking the artery.
- **Coronary artery spasm:** **Temporary narrowing** of coronary arteries due to muscle contraction.

Risk Factors

- **Hypertension** (high blood pressure)
- **Hyperlipidemia** (high cholesterol levels)
- **Smoking**
- **Diabetes mellitus**
- **Obesity**
- **Sedentary lifestyle**

- **Family history** of heart disease
- **Stress**
- **Age** (more common in older adults)
- **Male gender** is at higher risk.

Symptoms

- **Angina pectoris**: Chest pain or discomfort due to reduced blood flow to the heart.
- **Shortness of breath**: Difficulty breathing due to reduced heart function.
- **Fatigue**: Feeling tired easily.
- **Palpitations**: Sensation of a **rapid or irregular heartbeat**.
- **Nausea** and **dizziness** can also occur in severe cases.

Diagnosis

- **Clinical evaluation**: Based on symptoms and risk factors.
- **ECG (Electrocardiogram)**: Detects abnormal heart rhythms and signs of **heart ischemia**.
- **Echocardiogram**: Assesses heart function and **valve movement**.
- **Stress testing**: Monitors heart function during physical activity.
- **Coronary angiography**: To visualize the **coronary arteries** and identify blockages.
- **Blood tests**: Check for **cardiac markers** like **troponin** which increase after a heart attack.

Treatment

- **Lifestyle modification**:
 - Stop **smoking**.
 - Control **blood pressure** and **cholesterol**.
 - Regular **exercise**.
 - Maintain a **healthy diet**.
- **Medications**:
 - **Antiplatelets** (e.g., **aspirin**): Prevents **blood clots**.
 - **Beta-blockers**: Lower heart rate and blood pressure.
 - **Nitrates**: Relieves **chest pain** by relaxing coronary arteries.
 - **Calcium channel blockers**: Reduces heart workload.
 - **Statins**: Lower **cholesterol** levels and reduce plaque buildup.
 - **ACE inhibitors**: Lower blood pressure and protect heart function.

Surgical Treatment

- **Angioplasty**: A **balloon** is used to **open up narrowed arteries**.
- **Stent placement**: A small mesh tube is inserted to keep the artery open.

- **Coronary artery bypass grafting (CABG):** Surgical procedure to bypass blocked coronary arteries using a graft.

Prevention

- Control **risk factors** like **blood pressure**, **cholesterol**, and **blood sugar**.
- **Healthy lifestyle:** Balanced diet, regular exercise, weight control, and **stress management**.
- **Regular check-ups** for early detection of risk factors and treatment.

Q3. Cushing Syndrome

Causes

- **Excess cortisol** production by the adrenal glands.
- **Pituitary adenoma:** Overproduction of **ACTH** (adrenocorticotrophic hormone) leading to excess cortisol production by the adrenal glands.
- **Ectopic ACTH secretion:** Tumors outside the pituitary (e.g., lung cancer) produce ACTH, stimulating cortisol production.
- **Adrenal tumors:** Direct overproduction of cortisol by the adrenal glands.
- **Exogenous corticosteroid use:** Long-term use of **steroid medications** can cause **iatrogenic Cushing syndrome**.

Clinical Features

- **Obesity:** **Fat deposition** in the **face** (moon face), **upper back** (buffalo hump), and **abdomen** (central obesity).
- **Skin changes:** **Thin skin**, easy **bruising**, **purple striae** (stretch marks) on the abdomen, thighs, and breasts.
- **Hypertension:** **High blood pressure** due to increased cortisol levels.
- **Hyperglycemia:** Elevated **blood sugar** levels, leading to **diabetes mellitus** in some cases.
- **Muscle weakness:** Muscle **atrophy** and **weakness**, especially in the upper arms and thighs.
- **Osteoporosis:** **Weak bones** due to reduced calcium absorption and increased bone resorption.
- **Hirsutism:** **Excessive hair growth** in females, especially on the face and body.
- **Menstrual irregularities:** **Amenorrhea** or **oligomenorrhea** in females.
- **Psychiatric symptoms:** **Depression**, **irritability**, and **cognitive dysfunction**.
- **Increased susceptibility to infections:** Due to immune suppression by high cortisol levels.

Diagnosis

- **Clinical evaluation:** Based on symptoms and physical signs.
- **24-hour urinary cortisol excretion:** Increased cortisol in urine suggests Cushing syndrome.
- **Low-dose dexamethasone suppression test:** A **failure to suppress cortisol levels** after administering dexamethasone (a synthetic steroid) confirms the diagnosis.
- **ACTH levels:** Measurement of **serum ACTH** to differentiate between **ACTH-dependent** and **ACTH-independent** causes.
- **Imaging:**
 - **CT scan** or **MRI** to identify **pituitary tumors** or **adrenal tumors**.
 - **Chest X-ray** to check for **ectopic ACTH-producing tumors** (e.g., lung cancer).
- **Late-night salivary cortisol:** Elevated cortisol levels at night, when cortisol is normally low, suggest Cushing syndrome.

Treatment

- **Surgical treatment:**
 - **Pituitary surgery** to remove **adenomas** in ACTH-dependent cases.
 - **Adrenalectomy:** Removal of the adrenal glands in cases of adrenal tumors.
- **Medications:**
 - **Ketoconazole:** Reduces cortisol production in the adrenal glands.
 - **Metyrapone:** Inhibits cortisol synthesis.
 - **Mitotane:** Reduces adrenal cortex activity, used for adrenal tumors.
 - **Cabergoline:** In some cases of pituitary adenomas, this can lower ACTH secretion.
- **Radiation therapy:** For patients who cannot undergo surgery or when surgery is not successful.
- **Discontinuation of corticosteroid use:** Gradual tapering off of exogenous steroids if they are the cause of Cushing syndrome.

Q4. Describe clinical features, complications, diagnosis and management of chronic rheumatic mitral stenosis.

Mitral Stenosis (MS)

Causes

- **Rheumatic heart disease** is the most common cause.
- In **elderly**, heavy **calcification** of the mitral valve apparatus can be the cause.
- Rarely, it can be **congenital** or **degenerative**.
- More commonly seen in **females**.

Pathophysiology

- The normal mitral valve has an **orifice** of 4-6 cm².
- In **mitral stenosis**, the orifice narrows due to **fibrosis**, **calcification**, **fusion of commissures**, and **shortening of chordae tendineae**.
- **Orifice < 2 cm²** becomes **hemodynamically significant**, and **critical stenosis** occurs at **<1 cm²**.
- The obstruction increases **left atrial pressure**, causing **left atrial hypertrophy** and **dilatation**, leading to **pulmonary venous congestion** and **pulmonary hypertension**.
- **Right ventricular failure** can develop due to **pulmonary hypertension**.
- **Breathlessness** occurs due to reduced lung compliance and **low cardiac output**.
- **Atrial fibrillation (AF)** is common due to **left atrial dilatation**, leading to the risk of **left atrial thrombus** and **thromboembolism**.

Clinical Manifestations

- Early signs include **breathlessness on exertion** and **fatigue**.
- As the stenosis worsens, patients may experience **dyspnoea at rest**, **orthopnoea**, and **paroxysmal nocturnal dyspnoea (PND)**.
- **Acute pulmonary oedema** may occur.
- **Hemoptysis** (due to **pulmonary congestion**), **pulmonary embolism**, and **cough** due to **pulmonary congestion**.
- **Chest pain** due to **pulmonary venous hypertension**.
- Symptoms of **thromboembolic complications** like **stroke** and **limb ischaemia**.

Signs

- **Atrial fibrillation**.
- **Mitral facies** (flushed cheeks with a pale pallor).
- **Auscultation**:
 - **Loud first heart sound**, **opening snap**, and **mid-diastolic rumbling murmur** best heard at the **apex**.
- Signs of **pulmonary hypertension**:
 - **Right ventricular heave** and **loud P2**.
- Signs of **raised pulmonary capillary pressure**:
 - **Crepitations** (basal crackles), **pulmonary oedema**, **effusions**.
- **Ascites** and **pleural effusion** in severe cases.

Investigations

- **ECG**: Shows **LA enlargement**, **RV hypertrophy**, and **atrial fibrillation**.
- **Chest X-ray**: Reveals **LA enlargement** and **pulmonary congestion**.
- **Echocardiography**: Most **sensitive** and **specific** for diagnosing mitral stenosis. It detects **valvular abnormalities**, **cardiac chamber size**, **pulmonary artery pressure**, **ventricular dysfunction**, and **thrombi**.

- **Cardiac catheterization:** Assesses **associated valvular lesions** and detects **coronary artery disease**.

Treatment

A. Medical Management

- **Sodium restriction.**
- **Diuretics** to reduce **pulmonary congestion**.
- In **atrial fibrillation**, **digoxin** is given to control **ventricular rate**.
- **Beta-blockers** and **calcium antagonists** (e.g., **verapamil**, **diltiazem**) are used.
- **Oral anticoagulants** (**warfarin**) for patients with **thromboembolic history** or **AF**.
- **Prophylaxis** for **rheumatic fever**.

B. Surgical Management

- **Mitral valvotomy** when symptoms persist despite medical treatment or when stenosis is severe:
 - **Percutaneous balloon valvotomy** is done when the valve is **non-calcified** and without **regurgitation**.
 - **Open valvotomy** is done if balloon valvotomy is not possible or if restenosis occurs. The procedure loosens the **fusion of the valve** and removes **calcium deposits** and **thrombi**.
- **Mitral valve replacement:** In cases of **critical stenosis** or **significant mitral regurgitation**. Replacement is also done if the valve is **severely calcified** or **distorted**.

Q5. Describe aetiology, clinical profile, investigations and management of acute myocardial infarction.

Myocardial Infarction (MI)

Definition

- Necrosis of myocardium due to **imbalance between coronary blood supply and myocardium demand**.
- **Thrombus formation** at site of **plaque rupture**
- **Types:**
 - Subendocardial: Subendocardial region
 - Transmural: Full myocardial thickness
 - Silent: No symptoms, only ECG changes

Symptoms

- **Chest pain:** Severe, prolonged, tightness, heaviness
- **Breathlessness,** syncope, vomiting, fatigue
- **Anxiety,** fear of death
- Nausea, vomiting, collapse

Physical Signs

- **Sympathetic activation:** Pallor, sweating, tachycardia
- **Vagal activation:** Vomiting, bradycardia
- **Impaired myocardial function:** Hypotension, oliguria, cold extremities, raised JVP, lung crepitations
- **Tissue damage:** Mild fever, arrhythmias
- **Complications:** Mitral regurgitation, pericarditis, ventricular arrhythmias

Complications of MI

- **Cardiogenic shock:** Bad prognosis
- **Cardiac failure:** Pulmonary oedema
- **Papillary muscle infarction:** Mitral regurgitation
- **Septal rupture:** VSD murmur, hypotension
- **Cardiac tamponade:** Pericardial rupture
- **Embolism:** Cerebral, peripheral, DVT, PE
- **Dressler's syndrome:** Autoimmune post-MI reaction

Investigations

- **ECG:** ST elevation, Q waves, peaked T waves
- **Cardiac markers:**
 - CK-MB: Rises 4-6h, peaks 12h
 - Troponins: Sensitive, elevated 100-200h
- **Others:** Leucocytosis, raised ESR, CXR (pulmonary oedema), echocardiography (wall motion, EF)

Management

- **Immediate:** Oxygen, IV line, ECG monitoring, analgesia, **Aspirin 300 mg**
- **Reperfusion:** PCI (preferred), thrombolysis

Specific Therapy

- **Thrombolysis:** Streptokinase (1.5MU IV/1h), t-PA
- **Beta-blockers:** Reduce HR, BP, antiarrhythmic properties
- **ACE Inhibitors:** Prevent remodelling, start within 24h
- **Antiplatelets:** Aspirin 150 mg (Clopidogrel if intolerance)
- **Anticoagulants:** Heparin (DVT/PE prevention)

Thrombolysis Indications

- **ST elevation** >0.1mV, therapy ≤12h
- **Signs of reperfusion:** Chest pain relief, ST resolution, reperfusion arrhythmias

Complications of Thrombolysis

- **Bleeding** (puncture sites, brain), allergic reactions (STK), hypotension

Contraindications

- **Active bleeding**, recent surgery, **uncontrolled HTN**, pregnancy

PCI (Percutaneous Intervention)

- **Primary PTCA** within 90 min better than thrombolysis
- Use **Glycoprotein IIb/IIIa inhibitors** (Abciximab)

Right Ventricular Infarction

- **Hypotension** due to low RV stroke volume
- Treat: **Volume expansion**, avoid diuretics

Rehabilitation

- Gradual mobilization, return to work in **6 weeks**
- Exercise, smoking cessation, lipid control
- Prevent **arrhythmias**: Manage pain, correct hypokalemia

Q6. What are the symptoms and signs of congestive cardiac failure? How will you treat this condition.

Congestive Heart Failure (Cardiac Failure)

- Defined as a condition where ventricles at normal filling pressures cannot maintain an adequate cardiac output to meet metabolic needs or can only do so with elevated filling pressure.

Pathophysiology

- **Preload:** Pressure filling the left ventricle during diastole. Measured by left-ventricular end-diastolic pressure or pulmonary artery wedge pressure (PAWP).
- **Afterload:** Pressure against which left ventricle contracts. Measured by mean aortic pressure. Affected by total peripheral resistance and left ventricle size.
- **Myocardial contractility:** Depends on adrenergic activity and circulating catecholamines.

Heart Failure Characteristics

- **Decreased cardiac output:** Due to increased preload, increased afterload, and decreased myocardial contractility.
- **Compensatory mechanisms:** Initially beneficial, later counterproductive, mediated by the renin-angiotensin system and autonomic nervous system.

Compensatory Mechanisms

- **Increased myocardial contractility:** Ventricular dilation increases contractility, leading to a higher ejected blood volume (Frank-Starling law), but excessive stretch reduces contractility.
- **Myocardial hypertrophy:** Due to overload, it increases contractility but raises oxygen demand and reduces ventricular compliance.
- **Sympathetic stimulation:** Causes vasoconstriction, salt and water retention, increased heart rate, and contractility. Excessive afterload reduces cardiac output and increases oxygen demand.

Myocardial Remodelling

- Involves hypertrophy, apoptosis, and changes in extracellular matrix, leading to gradual loss of contractility.

Types of Heart Failure

- **Acute and Chronic Heart Failure:**
 - Acute: Sudden onset, causing hypoperfusion without edema (e.g., myocardial infarction).
 - Chronic: Gradual onset, maintaining blood pressure but with edema (e.g., dilated cardiomyopathy).
 - Compensated: No overt failure due to compensatory mechanisms.

- **High Output and Low Output Heart Failure:**
 - High Output: Increased cardiac output (e.g., hyperthyroidism, anemia).
 - Low Output: Low cardiac output (e.g., ischemic heart disease, hypertension).
- **Left-Sided, Right-Sided, and Biventricular Heart Failure:**
 - Left-sided: Fluid accumulates in lungs causing pulmonary congestion (e.g., myocardial infarction).
 - Right-sided: Fluid accumulates in systemic circulation causing venous congestion (e.g., cor pulmonale).
 - Biventricular: Failure of both ventricles (e.g., dilated cardiomyopathy).
- **Forward and Backward Heart Failure:**
 - Forward: Decreased cardiac output.
 - Backward: Normal output with congestion due to salt and water retention.
- **Systolic and Diastolic Heart Failure:**
 - Systolic: Abnormal contraction (ejection fraction < 40%).
 - Diastolic: Abnormal relaxation.

Aetiology Congestive Heart Failure (Cardiac Failure)

- **Pressure Overload:**
 - Systemic hypertension
 - Pulmonary hypertension
 - Aortic stenosis
 - Pulmonary stenosis
- **Volume Overload:**
 - Mitral regurgitation
 - Aortic regurgitation
 - Ventricular septal defect
 - Persistent ductus arteriosus
 - Atrial septal defect
- **Impaired Ventricular Function:**
 - Myocarditis
 - Cardiomyopathy
 - Myocardial infarction

Clinical Manifestations

- **Dyspnoea:** Breathlessness on exertion, worsening over time.
- **Orthopnoea:** Dyspnoea when lying flat, relieved by sitting up.
- **Paroxysmal Nocturnal Dyspnoea (PND):** Severe shortness of breath at night, not relieved by sitting.
- **Cardiac Asthma:** Wheezing, cough, and dyspnoea at night.
- **Acute Pulmonary Oedema:** Severe breathlessness with pink frothy sputum and crepitations.
- **Cheyne-Stokes Respiration:** Periodic breathing with apnoea and hyperventilation.

- **Nocturia:** Increased urine output at night.
- **Cerebral Symptoms:** Confusion, memory impairment, anxiety.
- **Nonspecific Symptoms:** Fatigue, anorexia, nausea, abdominal fullness.
- **Cardiac Oedema:** Dependent oedema, most noticeable in legs, less in the morning.
- **Cyanosis:** Cold, pale extremities with blue lips/nail beds.
- **Pulse:** Pulsus alternans in severe heart failure.
- **Blood Pressure:** Diminished pulse pressure and possible hypotension in acute failure.
- **Heart Sounds:** S3 and S4 gallop sounds suggestive of heart failure.
- **Respiratory:** Dull percussion and inspiratory crepitations in lung bases.
- **Liver:** Hepatomegaly, right upper quadrant pain, possible jaundice.
- **Kidney:** Oliguria, elevated blood urea (prerenal azotemia).
- **Cardiac Cachexia:** Severe weight loss and malnutrition.

Diagnostic Studies

- **Chest Radiograph:** Cardiomegaly, pulmonary oedema, Kerley lines.
- **ECG:** For arrhythmias and ischemic changes.
- **Echocardiography:** For ejection fraction and valvular function.
- **BNP Levels:** Elevated in heart failure.
- **Renal and Liver Function Tests:** To monitor kidney and liver status.

Management of Heart Failure

- **General Measures:**
 - Physical and emotional rest to reduce oxygen demand.
 - Weight management and salt restriction.
- **Drug Therapy:**
 - **Diuretics:** For fluid removal (e.g., furosemide, spironolactone).
 - **Vasodilators:** ACE inhibitors, ARBs, nitrates, hydralazine.
 - **Inotropic Agents:** Digitalis, dopamine, dobutamine.
 - **Beta Blockers:** Metoprolol, carvedilol.

Treatment aims to reduce preload, afterload, and improve myocardial contractility using a combination of drugs.

Q7. Describe clinical features, investigations and treatment of Acute Rheumatic Fever (ARF)

- **ARF** is a multisystem disorder that follows **pharyngeal streptococcal infection**.
- It is the **most common cause of acquired heart disease in children and adolescents**.
- Diagnosis requires **2 or more major criteria**, or **1 major and 2 minor criteria**, along with evidence of **previous streptococcal infection**.

Common Symptoms

- **Sore throat:** Only **2/3 of patients** recall upper respiratory symptoms in the past **1-5 weeks**.
- **Polyarthrititis:** **75%** of cases, involving **large joints** like **ankles, knees, elbows**. The pain is **migratory**.
- **Carditis:** **40-50%** of cases, involving **pericardium, endocardium, and myocardium**. Can lead to **mitral regurgitation**.
- **Chorea:** Involuntary movements of hands, feet, or face. Appears **6 months after** streptococcal infection.
- **Subcutaneous nodules:** Small, **painless** bumps, often associated with **severe carditis**.
- **Erythema marginatum:** Transient **rash** that appears as **macules** with **clear center** and **serpiginous margins**.

Pathogenesis

- **ARF** affects children aged **5-15 years**.
- It occurs in about **3%** of those with **streptococcal pharyngitis**.
- **Streptococcal antigens** cross-react with **cardiac tissue**, triggering **autoimmune** responses.

Diagnosis (Jones's Criteria)

- **Major criteria:**
 - **Polyarthrititis**
 - **Carditis**
 - **Chorea**
 - **Subcutaneous nodules**
 - **Erythema marginatum**
- **Minor criteria:**
 - **Fever, arthralgia**
 - **Raised ESR, CRP**
 - **Prolonged PR interval**
 - Positive throat culture for **streptococci** or elevated **streptococcal antibody titers**.

Clinical Manifestations

- **Fever, lethargy, anorexia**
- **Carditis:** Tachycardia, murmur of **mitral regurgitation**, **cardiomegaly**.
- **Chorea:** **Sydenham's chorea** or **St. Vitus dance**. Involuntary movements of face, hands, and feet.

Laboratory Features

- **Leucocytosis, raised ESR**
- **Throat swab** may show **group A streptococci**
- **ASO titer** (>200 units in adults, >300 units in children)
- **ASTZ test** for **recent streptococcal infection**
- **ECG: Prolonged PR interval**

Management

1. **Bed Rest:**
 - For **non-carditis**: Rest until **ESR** and **temperature** normalize.
 - For **carditis**: Rest for **2-6 weeks** after normalization.
2. **Antistreptococcal Therapy:**
 - **Benzathine penicillin 1.2 million** units IM.
 - Or **oral erythromycin** (if allergic to penicillin).
3. **Salicylates (Aspirin):**
 - **60-120 mg/kg/day** for arthritis.
 - Taper as ESR normalizes.
4. **Corticosteroids:**
 - **Prednisolone** for severe **carditis** or **arthritis**.
 - Taper and overlap with **aspirin** to avoid **rebound** effects.
5. **Supportive Therapy:**
 - Treat **congestive heart failure**, **valvular lesions**, and **chorea**.

Prevention

1. **Primary Prevention:**
 - **Accurate diagnosis** and **treatment** of **group A streptococcal pharyngitis**.
 - Mass **penicillin prophylaxis** in closed outbreaks.
2. **Secondary Prevention:**
 - Continuous prophylaxis for those with **previous rheumatic fever**.
 - **Penicillin** every **3 weeks** or **oral erythromycin** if allergic.

Q8. Disseminated Intravascular Coagulation (DIC)

Definition

- **DIC** is a **serious disorder** characterized by **widespread activation of coagulation** in the blood vessels. This leads to the formation of **small blood clots** throughout the body, consuming clotting factors and platelets, which results in **bleeding** and **organ damage**.

Causes Disseminated Intravascular Coagulation (DIC)

- **Infections** (sepsis, particularly with **gram-negative bacteria**)
- **Trauma** or **surgery**
- **Obstetric complications** (e.g., **placental abruption, eclampsia**)
- **Cancer** (particularly **leukemia, solid tumors**)
- **Severe burns**
- **Shock** (hypovolemic, septic)
- **Transfusion reactions**

Pathophysiology

- In DIC, **endothelial injury** leads to the activation of the **coagulation cascade**, causing **excessive clotting** in the small blood vessels.
- **Platelets** and **clotting factors** are consumed, leading to **deficiency** in the bloodstream, causing **bleeding**.
- The formation of **microthrombi** obstructs blood flow to organs, leading to **ischemia** and **organ dysfunction**.

Clinical Features

1. **Bleeding:**
 - **Gum bleeding, nosebleeds, petechiae, purpura**
 - **GI bleeding** (e.g., **melena** or **hematemesis**)
 - **Prolonged bleeding** after injury or surgery
2. **Organ Dysfunction:**
 - **Renal failure** (due to **microthrombi** in the kidneys)
 - **Liver dysfunction** (elevated **liver enzymes**)
 - **Respiratory failure** (due to **pulmonary microthrombi**)
3. **Thrombotic Features:**
 - **Gangrene** or **limb ischemia**
 - **Cerebrovascular accidents** (due to clot formation in the brain)

Diagnosis

- **Blood tests:**
 - **Prolonged PT** and **aPTT**
 - **Low platelets**
 - **Low fibrinogen levels**
 - **Increased D-dimer** (a marker of clot breakdown)
 - **Schistocytes** (fragmented red blood cells on peripheral smear)
- **Coagulation tests:**
 - **Positive fibrin degradation products (FDPs)**
 - **Elevated thrombin time**

Management

1. **Treat Underlying Cause:**
 - **Infections:** Antibiotics
 - **Obstetric complications:** Manage pregnancy-related issues
 - **Trauma or surgery:** Control bleeding and prevent further injury
2. **Supportive Treatment:**
 - **Platelet transfusion** if **platelets** are low
 - **Fresh frozen plasma (FFP)** to replenish **coagulation factors**
 - **Cryoprecipitate** if **fibrinogen** levels are low
3. **Anticoagulation:**
 - **Heparin** may be used in certain cases of DIC to prevent further clotting, but **caution** is needed due to the risk of bleeding.
4. **Organ Support:**
 - **Renal dialysis** if **renal failure** occurs
 - **Ventilator support** for **respiratory failure**

Prognosis

- The **outcome** of DIC depends on the **underlying cause** and how quickly treatment is initiated.
- **Prompt identification and treatment** are crucial for improving survival rates.

Q9. Name Anticoagulant Drugs, Uses, and Precautions in Dental Practice

1. Warfarin

- **Use:**
 - Prevents **thrombosis** in conditions like **atrial fibrillation**, **deep vein thrombosis (DVT)**, and **pulmonary embolism (PE)**.
 - Used in patients with **mechanical heart valves**.
- **Precautions in Dental Practice:**
 - **Monitor INR** (International Normalized Ratio) to ensure **safe clotting levels**.
 - Avoid **invasive dental procedures** if INR is high (greater than 3).
 - Ensure **good oral hygiene** to avoid **gingival bleeding**.
 - Avoid **prolonged bleeding** post-extraction by using **pressure packs**.
 - Consider **local hemostatic measures** (e.g., **gel foam**, **hemostatic sponges**) after surgery.

2. Heparin

- **Use:**
 - Used in hospital settings for **acute thrombosis**, **myocardial infarction**, and **post-surgical** patients.
 - **Low molecular weight heparin** (e.g., **enoxaparin**) is often used in outpatient settings.

- **Precautions in Dental Practice:**
 - If patients are on **heparin therapy**, consult with their **physician** about stopping it prior to dental procedures.
 - **Monitor for bleeding complications**, especially in invasive procedures like extractions.
 - **Avoid trauma** to mucous membranes to reduce the risk of excessive bleeding.

3. Direct Oral Anticoagulants (DOACs)

- **Examples: Apixaban, Rivaroxaban, Dabigatran**
- **Use:**
 - Used for prevention and treatment of **DVT, PE**, and **atrial fibrillation**.
 - Often preferred due to **no routine monitoring** required (e.g., **INR** is not needed).
- **Precautions in Dental Practice:**
 - **Timing of last dose:** Dental procedures should be scheduled at least **12-24 hours** after the last dose to minimize bleeding risk.
 - **Monitor for prolonged bleeding** post-extraction and use **local hemostatic agents** as needed.
 - Consult with the physician if the patient is on **Dabigatran** and requires a procedure, as it can have a longer half-life.

4. Aspirin

- **Use:**
 - Used as a **blood thinner** in patients at risk for **heart attacks** and **strokes**.
 - **Low-dose aspirin** is often used for **cardiovascular disease prevention**.
- **Precautions in Dental Practice:**
 - **Aspirin** is a **platelet inhibitor** and may increase bleeding during dental procedures, especially after **extractions**.
 - **Discontinue aspirin 7 days before** elective dental surgery** to reduce bleeding risk.
 - For **high-risk cardiac patients**, the benefit of continuing aspirin often outweighs the risk of bleeding, so **consult with the physician**.

5. Clopidogrel (Plavix)

- **Use:**
 - Used in patients who have had a **stent placement, myocardial infarction**, or **stroke** to prevent **thrombosis**.
- **Precautions in Dental Practice:**
 - **Avoid invasive procedures** immediately after starting Clopidogrel, as it **inhibits platelet aggregation**.

- **Consult with the physician** about stopping Clopidogrel prior to dental work, especially for procedures like **oral surgery** or **extractions**.
- **Monitor for bleeding**, especially in the case of **tooth extractions** and **post-operative care**.

6. Dipyridamole

- **Use:**
 - Often combined with **aspirin** for **stroke prevention** and in patients with **prosthetic heart valves**.
- **Precautions in Dental Practice:**
 - **Bleeding risk** is increased with **dipyridamole**, especially in invasive procedures.
 - **Monitor bleeding times** and apply appropriate **hemostatic measures** post-surgery.

General Precautions in Dental Practice for Anticoagulants

- **Thorough medical history:** Always ask the patient about anticoagulant therapy during **pre-treatment evaluation**.
- **Consult with physician:** In cases of **high-risk patients** on anticoagulants, always **consult with their physician** before performing invasive procedures.
- **Local hemostatic measures:** Use measures like **pressure, sutures**, and **hemostatic agents** (e.g., **gelatin sponges, bone wax**) to control bleeding.
- **Post-procedure care:** Advise patients to **avoid excessive physical activity** post-surgery to minimize the risk of bleeding.

Q10. Describe signs and symptoms of angina and how you would manage the case.

- Clinical syndrome of **discomfort due to transient myocardial ischemia**.
- Caused by:
 - **Obstruction of coronary flow by atheroma.**
 - **Coronary arterial spasm.**
- Worsened by factors that increase **myocardial oxygen demand** or reduce supply:
 - **Exercise, Anaemia, Tachycardia, Hypertension, Hyperthyroidism, Aortic stenosis, Aortic regurgitation, Arrhythmias.**

Clinical Features

- **Pain:** Retro-sternal, triggered by exertion, relieved by **rest or sublingual nitrates**.
- **Character:** Squeezing, crushing, aching. Radiates to **left arm**, sometimes to right arm, throat, back, chin, or epigastrium.
- **Angina decubitus:** Pain while lying flat.

- **Nocturnal angina:** Unusual, occurs in **aortic regurgitation**, associated with **nightmares, sweating, wide pulse pressure**, does not respond to nitrates.
- **Prinzmetal's angina:** Due to **coronary artery spasm**, transient **ST elevation** on ECG.

Examination

- **Physical signs during pain:**
 - Rise in BP and heart rate.
 - Fourth heart sound.
 - Mitral regurgitation murmur.
 - Paradoxical splitting of second heart sound.

Investigations

- **ECG:** Reversible **ST depression/elevation**, may require **treadmill testing**.
- **Myocardial perfusion scan:** Using **radioactive thallium**.
- **Coronary arteriography:** Identifies extent and site of stenosis.

Management

- **General Measures:**
 - Explain disease properly to patient.
 - Avoid walking after meals, cold weather, or strenuous exercise.
 - Treat **hyperlipidemia, hypertension, diabetes**.
 - Correct **anaemia, valvular diseases, arrhythmias**.

Drug Treatment

- **Nitrates:**
 - **Glyceryl trinitrate** (sublingual) relieves pain in **2-3 minutes**.
 - **Isosorbide dinitrate/mononitrate** used for **prolonged action**, ensure **nitrate-free period** to prevent tolerance.
- **Beta-blockers:**
 - Reduce **myocardial oxygen demand**, heart rate, and anxiety response.
 - **Propranolol, Metoprolol, Atenolol** commonly used.
 - Avoid abrupt withdrawal.
- **Calcium antagonists:**
 - Cause **vasodilation** and reduce **cardiac excitability**.
 - Examples: **Nifedipine, Verapamil, Diltiazem**.
- **Antiplatelet agents:**
 - **Aspirin** (75-150 mg/day) inhibits **thromboxane A2**.
 - **Clopidogrel** for patients intolerant to aspirin.

Surgical Treatment

- **CABG:** Saphenous vein or internal mammary artery graft for severe **stenosis**.
- **PTCA:** Balloon dilatation, often with **drug-coated stents** to prevent re-occlusion.

Q11. Discuss aetiology and clinical features of subacute bacterial endocarditis. Add a note on principles of management of Infective Endocarditis prophylaxis.

Microbial infection of heart endothelium, commonly affecting heart valves or endocardium of chambers.

- In arterial sites like **AV shunts, PDA, or coarctation of aorta**, termed **infective endarteritis**.

Pathology

- Damaged **endothelium** (e.g., high-pressure jet injury) leads to platelet-fibrin deposition (**non-bacterial thrombotic endocarditis**).
- Microbes enter bloodstream (via mucosa, skin, infections), colonize deposits, forming **vegetations**.
- **Common organisms:**
 - **Subacute:** *Viridans streptococci* (from oral flora).
 - **Acute:** *Staphylococcus aureus*.
 - **Prosthetic valves:** *Staph. epidermidis*, **HACEK group**, fungi, anaerobes.

Clinical Features

- **General:** Fever, weight loss, night sweats, weakness.
- **Cardiac:** New murmur, heart failure, heart blocks.
- **Extracardiac:**
 - **Signs:** Anaemia, clubbing, splenomegaly, petechiae, **Janeway lesions, Osler nodes, Roth spots**.
 - **Neurological:** Meningitis, embolic infarcts, mycotic aneurysm rupture.
 - **Other:** Glomerulonephritis, septic emboli, organ infarcts.

Investigations

- **Blood culture:** Identifies organism, guides therapy.
- **Echocardiography:** Detects vegetations, abscess, heart function.
 - **Transoesophageal echocardiography:** More sensitive.
- Other findings: Anaemia, leukocytosis, high ESR/CRP, haematuria.

Duke's Criteria for Diagnosis

- **Major:** Positive blood culture, echocardiographic findings, new regurgitation.

- **Minor:** Fever, embolic/vasculitic events, predisposing conditions, suggestive findings.
- **Definite IE:** 2 major or 1 major + 3 minor.
- **Possible IE:** 1 major + 1 minor or 3 minor.

Treatment Principles

- **Antibiotics:**
 - Must be **bactericidal**, parenteral, prolonged duration.
 - Based on culture sensitivity and **MIC values**.
- **Empirical Therapy:**
 - Subacute: **Ceftriaxone + gentamicin**.
 - Prosthetic valves: Add **vancomycin**.
- Common antibiotics:
 - Penicillin G, Gentamicin, Ceftriaxone, Ampicillin, Vancomycin.

Surgical Indications

- Needed for **prosthetic valve**, fungal endocarditis, **heart failure**, abscess, large vegetations.

Prophylaxis

- High-risk patients (valvular/congenital heart diseases) before **bacteraemia-inducing procedures**.
- **Antibiotic Regimen:**
 - Standard: **Amoxicillin 2g oral**, 1 hr before.
 - Penicillin allergy: **Clarithromycin 500mg** or **Clindamycin 600mg oral**.
 - Unable to take oral meds: **Ampicillin IV** or **Clindamycin IV**.

Q12. Cyanosis

- **Definition:** Bluish discoloration of skin and mucous membranes due to increased reduced haemoglobin (>5 g/dl) or abnormal haemoglobin derivatives in capillary blood.
- **Common areas:** Most marked in **lips, nail beds, ears**, and **malar eminences**.

Types of Cyanosis

- **Central Cyanosis:** Caused by decreased **arterial oxygen saturation** or presence of abnormal haemoglobin derivatives.
- **Peripheral Cyanosis:** Due to diminished **peripheral blood flow**, often from reduced **cardiac output** or local vasoconstriction.

Causes of Central Cyanosis

- **Decreased arterial oxygen saturation:**
 - **Cardiac causes:**
 - **Fallot's Tetralogy:** A congenital heart defect causing mixed oxygenated and deoxygenated blood.
 - **Eisenmenger's Syndrome:** Reversal of shunt due to long-standing congenital heart defects leading to pulmonary hypertension.
 - **Congestive cardiac failure:** Reduced cardiac output leads to insufficient oxygenation of blood.
 - **Pulmonary causes:**
 - **Chronic bronchitis:** Obstructs airflow, reducing oxygen levels in blood.
 - **Interstitial lung disease:** Impairs gas exchange in lungs.
 - **Pulmonary arteriovenous fistula:** Abnormal connection between pulmonary arteries and veins bypassing gas exchange.
 - **High altitude:** Lower oxygen levels at high altitudes reduce blood oxygen saturation.
 - **Cirrhosis of liver:** May lead to pulmonary vasodilation, impairing oxygenation.
- **Abnormal haemoglobin derivatives:**
 - **Methaemoglobinaemia:** Hemoglobin is unable to bind oxygen effectively.
 - **Sulphaemoglobinaemia:** Presence of sulphur in hemoglobin reduces its oxygen-carrying capacity.

Causes of Peripheral Cyanosis

- **Reduced cardiac output:**
 - **Mitral stenosis:** Impairs blood flow from the left atrium to left ventricle, reducing oxygenated blood flow.
 - **Shock:** Severe circulatory failure causing low blood flow to extremities.
 - **Congestive cardiac failure:** Inadequate pumping by the heart, leading to poor circulation to the periphery.
- **Local vasoconstriction:**
 - **Cold exposure:** Causes vasoconstriction in extremities, reducing blood flow.
 - **Peripheral vascular disease:** Narrowing of blood vessels in the limbs, leading to reduced blood flow.

Q13. Syncope (Fainting)

- **Definition:** Generalized weakness of muscles, loss of postural tone, inability to stand, and loss of consciousness.
- **Faintness:** A prodromal phase of syncope, with symptoms like light-headedness, blurring of vision, heaviness in the lower limbs, and eventual loss of consciousness.

Mechanism of Syncope

- **Fainting** is due to **cerebral ischemia**, especially affecting the **brainstem**.
- **Faintness** occurs due to insufficient ischemia to impair consciousness. Causes include sudden **vasodilatation**, a **sudden fall in cardiac output**, or both.

Types of Syncope

1. Vasovagal Syncope:

- **Mechanism:** Reflex slowing of the heart through the **vagus nerve**, with a fall in **arterial pressure** and **peripheral vascular resistance**.
- **Causes:** Emotional stress, warm overcrowded rooms, sudden pain, mild blood loss, anemia, fever, and fasting.
- **Clinical features:**
 - Prodromal phase: **nausea, sweating, yawning, epigastric distress, tachypnoea**, weakness, and confusion.
 - Followed by: **faintness, pallor, coldness of hands and feet**, and loss of consciousness.
- **Examination:**
 - Tachycardia in the prodromal phase, then **bradycardia, low blood pressure**, and **weak pulse**.
 - **Consciousness is regained rapidly**.
- **Common in young people.**

2. Cardiac Syncope:

- **Cause:** Sudden reduction in cardiac output, can be due to the heart beating too fast, too slow, or not at all.
- **Common causes:**
 - **Complete heart block** (Stokes-Adams-Morgagni syndrome).
 - **Paroxysmal tachycardias** (supraventricular or ventricular).
 - **Sick sinus syndrome**.
 - **Acute massive myocardial infarction**.
 - **Aortic stenosis** (exertional syncope).
 - **Hypertrophic cardiomyopathy** (exertional syncope).
 - **Primary pulmonary hypertension**.
 - **Pulmonary embolism**.
 - **Left atrial ball valve thrombus or myxoma**.
 - **Tetralogy of Fallot**.

3. Orthostatic Syncope (Postural Syncope):

- **Cause:** Occurs when standing up quickly or standing still for long, leading to **postural hypotension** due to loss of vasoconstrictor reflexes in the lower limb vessels.
- **Common causes:**
 - **Physiological**.
 - **Idiopathic orthostatic hypotension**.
 - **Drugs:** Antihypertensive drugs, **ACE inhibitors, beta-blockers, calcium channel blockers, diuretics, nitrates**.

- **Diabetic neuropathy** and **tabes dorsalis**.
 - **Haemorrhage**.
 - **Vasodilators** and **volume depletion**.
4. **Syncope Associated with Cerebrovascular Disease:**
- **Cause: Occlusion** in large neck arteries reduces blood flow to the brainstem, especially during physical activity, causing loss of consciousness.
5. **Miscellaneous Causes of Syncope:**
- **Carotid Sinus Syncope:** Compression of the **carotid sinus** (e.g., turning the head, tight collar, shaving) in elderly patients.
 - **Vagal and glossopharyngeal neuralgia**.
 - **Micturition syncope:** Occurs during or after micturition in elderly, especially after rising from bed.
 - **Cough syncope:** Seen in elderly with chronic bronchitis after paroxysms of cough.
 - **Hypoglycaemia**.
 - **Hyperventilation** and **hysterical fainting**.

Q14. CPR

- Provides artificial **ventilation** and **perfusion** to vital organs (heart, brain) until normal cardiopulmonary function is restored.
- Includes **Basic Life Support (BLS)** and **Advanced Life Support (ALS)**.
 - **BLS:** Maintains oxygen and perfusion to vital organs until ALS is available.
 - **ALS:** Restores normal cardiac rhythm or cardiac output.

Phases of Life Support

- **Phase 1: BLS** - Circulation, Airway, Breathing (CAB).
- **Phase 2: ALS** - Drugs, ECG, Fibrillation.
- **Phase 3: Prolonged Life Support** - Post-resuscitation care.

Change in CPR sequence:

- From **ABC** (Airway, Breathing, Circulation) to **CAB** (Circulation, Airway, Breathing).

Circulation

- Start **chest compressions** immediately in a patient without central pulse.
- **Chest compressions:**
 - Place the patient on a hard surface.
 - Hands positioned on the lower half of the sternum, **2 fingers above the xiphoid process**.
 - Depress sternum **5 cm in adults, 2-4 cm in children, 1-2 cm in infants**.
 - Rate: **100-120/min**.

- Ratio: **30:2 (single rescuer), 30:2 (two rescuers)** without interruption for ventilation.
- Reassess rhythm every **2 minutes (5 cycles)**.

Airway Management

- **Basic techniques:**
 - **Head tilt, chin lift:** Displaces tongue anteriorly.
 - **Jaw thrust:** Lifts mandible forward.
 - **Finger sweep:** Clears foreign body (only in unconscious patients).
 - **Heimlich maneuver:** Removes foreign body by abdominal thrusts.

Breathing

- **Mouth-to-mouth breathing:** Seal lips, blow air steadily, and watch for chest rise.
- **Mouth-to-nose breathing:** Used in case of mouth injury.
- **Infants:** Use **mouth-to-mouth and nose** technique.

Termination of BLS

- Pulse and respiration return.
- Medical help arrives or rescuer becomes exhausted.

ALS Techniques

- **Airway management:** Includes **endotracheal intubation, cricothyrotomy, tracheostomy**.
- **Advanced breathing:** Use of **Ambu bag**, 100% oxygen, or ventilator.
- **Advanced circulation:** Chest compression + IV access, defibrillation, and rhythm assessment.

Defibrillation

- Used for **shockable rhythms** (Ventricular Tachycardia, Ventricular Fibrillation).
- Voltage: **150-360 J biphasic** or **360 J monophasic**.
- CPR continues for 2 minutes after each shock.

Drugs in ALS

- **Adrenaline:** 1 mg IV every 3-5 minutes.
- **Amiodarone:** 300 mg IV after the 3rd shock.
- **Magnesium:** 2 g IV for Torsade de Pointes, digoxin toxicity.
- **Calcium:** 10 mL of 10% calcium chloride IV (used in hyperkalemia, hypocalcemia).
- **Sodium bicarbonate:** For severe metabolic acidosis or hyperkalemia.

Post-Resuscitation Care

- Maintain **airway and breathing**.
- Continue cardiac monitoring.
- Use **vasoactive drugs** (norepinephrine, dobutamine).
- Avoid **hyperthermia** and **hyperglycemia (<200 mg/dL)**.
- Treat underlying cause of cardiac arrest.